

SRINIVAS UNIVERSITY
INSTITUTE OF COMPUTER SCIENCE AND INFORMATION SCIENCES
I SEMESTER MCA- 2025

COURSE CODE: **25MCA11**; COURSE NAME: **LINUX SYSTEM ADMINISTRATION**; CREDITS: **4**

Max. Marks: 50

Time: 2 hours

MCA Program Outcomes (POs)

- PO1. Foundational Knowledge** – Demonstrate knowledge of statistics, computer science, and data science fundamentals necessary for the solution of real-world problems.
- PO2. Problem Analysis** – Identify, analyze, and interpret data-centric problems using scientific and computational thinking.
- PO3. Design & Development of Solutions** – Apply modern tools, programming, and data-driven techniques to design and develop efficient solutions, predictive models, and decision systems.
- PO4. Modern Tool Usage** – Create, select, and apply appropriate techniques, resources, and modern data science and IT tools such as Python, ML frameworks, and big data platforms.
- PO5. Data Interpretation & Analytics** – Apply statistical, machine learning, and AI techniques to discover meaningful insights from structured and unstructured datasets.
- PO6. Communication & Teamwork** – Communicate effectively with peers, industry professionals, and society at large, and function effectively as an individual or as part of diverse teams.
- PO7. Professional Ethics & Responsibility** – Recognize the need for Professional Ethics, Data Privacy, and responsibility in data-driven decision-making.

PO8. Indian Knowledge System (IKS) - To impart Indian Knowledge System among student community.

PO9. Innovations and Research – Use research-based knowledge and methods, including data collection, experimentation, hypothesis testing, and interpretation to derive valid conclusions.

PO10. Life-long Learning – Recognize the need for and prepare for independent and life-long learning to adapt to emerging technologies and global challenges.

PO11. Employability & Entrepreneurship – Acquire domain-specific knowledge and industry-oriented skills to pursue careers in data science, analytics, AI, and related fields and to develop an entrepreneurial mindset with the ability to identify opportunities, innovate solutions, and create sustainable ventures by applying technical, managerial, and leadership skills.

PO12. Health and wellness, Human Values & Social Responsibility – Develop awareness of physical and mental well-being, uphold ethical and human values, and contribute responsibly to society and the environment.

Course Outcomes:

CO1: to compare and contrast various Linux distributions and install and configure, managing file and directory permissions

CO2: to effectively manage users and groups, delegate administrative privileges securely, and configure authentication mechanisms

CO3: to use of yum for package management, including installing, updating, and removing software packages, as well as managing services with systemd and ensuring the system is secure and up-to-date.

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Module 1

Sl No	Question	Marks	Type	CO	PO
1	Why the name Linux has been derived?	1	K	C1	P4
2	What is a notation used for Current Directory in Linux?	1	K	C2	P4
3	What is a notation used for Parent Directory in Linux?	1	K	C2	P4
4	What is a significance of /dev directory?	1	U	C2	P4
5	What is a significance of /bin directory?	1	U	C2	P4
6	What is a significance of /etc directory?	1	U	C2	P4
7	What is the difference between cp and mv command?	1	S	C3	P3
8	What is the use of chmod command?	1	S	C3	P3
9	What is the use of chown command?	1	S	C3	P3
1	Describe the file system structure in Linux.	7	K	C2	P4
2	Give the features of Linux O.S	7	K	C1	P4
3	Give the structure of Linux OS with a diagram?	7	K	C1	P4
4	What is the difference between naming the file with absolute path and relative path? Illustrate with an example.	7	U	C2	P4
5	Give the different levels of security implementation available with Linux O.S.	7	U	C2	P3
6	List the different operations and their equivalent decimal values on a file for different users in Linux	7	U	C3	P3
7	Illustrate how to change different file permissions in Linux with example.	7	S	C3	P3
8	Give the usage of chown and chmod commands in Linux?	7	S	C3	P3
9	Illustrate how to change ownership of a file in Linux with example.	7	S	C3	P3
1	In detail give the structure of Linux OS with various components and their interactions	10	S	C1	P4
2	Illustrate with examples how to change ownership as well as permissions of a file.	10	S	C3	P3

Module 2

Sl No	Question	Marks	Type	CO	PO
1	Name the resource to find compatibility list with RedHat Linux	1	K	C1	P4
2	MBR stands for ---	1	K	C2	P4
3	GRUB stands for ---	1	K	C2	P4
4	Give the significance of Boot Loader	1	U	C2	P4
5	Give the significance of LILO	1	U	C2	P4
6	Give the Device Name assigned for the 1st logical partition in IDE	1	U	C2	P4
7	Give the significance Mounting a File System	1	S	C3	P3
8	Give the significance a Firewall	1	S	C3	P3
9	Name the default Boot Loader in Linux	1	S	C3	P3
1	Give the different alternate approaches for installing RedHat Linux on your system	7	K	C2	P4
2	Why the hardware issues to consider with importance while installing RedHat Linux?	7	K	C1	P4
3	Give the Disk structure and Partitioning approaches in RedHat Linux	7	K	C1	P4
4	What is MBR? Give its contents with its significance	7	U	C2	P4
5	What is Disk Druid (DD)? Illustrate managing the partitions using DD	7	U	C2	P3
6	What are the minimum partition required to install a Linux? Give their significance.	7	U	C3	P3
7	Illustrate a typical File System Mounting on different partitions in Linux	7	S	C3	P3
8	Illustrate different flavours of RedHat Linux that could be installed during Installation	7	S	C3	P3
9	Illustrate a typical Booting Process in any systems	7	S	C3	P3
1	Illustrate a general approach of assigning Device Name for different partition with example	10	S	C1	P4
2	Analyze various flavours of RedHat Linux that could be installed during its installation	10	S	C3	P3

Module 3

Sl No	Question	Marks	Type	CO	PO
1	What is the concept of a group in RedHat Linux?	1	K	C1	P4
2	In Linux, UID stands for what?	1	K	C2	P4
3	Name the admin tool for registering a RedHat Linux	1	K	C2	P4
4	What is the use of sudo command in RedHat Linux?	1	U	C2	P4
5	Identify the default administrate group in a RedHat Linux	1	U	C2	P4
6	What is the significance of baseurl in repos configuration file?	1	U	C2	P4
7	How can you login as a different user in RedHat Linux?	1	S	C3	P3
8	How can you add user to a existing group in RedHat Linux?	1	S	C3	P3
9	How to find the attached subscriptions in RedHat Linux?	1	S	C3	P3
1	What is a shadow password? What are the advantages with it?	7	K	C2	P4
2	What is a significance of group directories in RedHat Linux? Explain	7	K	C1	P4
3	What is the yum tool in RedHat Linux? Give its significance	7	K	C1	P4
4	Give the procedure of creating a group and adding a user to that group in RedHat Linux	7	U	C2	P4
5	How the sudo command is useful in RedHat Linux Management?	7	U	C2	P3
6	How the subscription-manager tool can be used in RedHat Linux Management?	7	U	C3	P3
7	Give the procedure to restrict the access of su command only to few users	7	S	C3	P3
8	Give the procedure to register as well as subscribe appropriate Red Hat enterprise server	7	S	C3	P3
9	Give the procedure to update your RedHat Linux system in offline mode	7	S	C3	P3
1	Demonstrate the procedure to create a Group Directory and set its GID, so that all group users can work on its files.	10	S	C1	P4
2	Give the usage of subscription-manager tool while administering RedHat Linux	10	S	C3	P3

Module 4

Sl No	Question	Marks	Type	CO	PO
1	Name the tool for package management in RedHat Linux	1	K	C1	P4
2	Name the tool for system and service manager in RedHat Linux	1	K	C2	P4
3	Give the command to get the status of httpd.service	1	K	C2	P4
4	What is systemd in RedHat Linux	1	U	C2	P4
5	How do you identify the unit files related to any Service?	1	U	C2	P4
6	Which target unit to be loaded for RedHat Linux to work in Graphical Mode?	1	U	C2	P4
7	How to get the Default Target Unit in RedHat Linux?	1	S	C3	P3
8	How to set RedHat Linux to load in graphical mode by default	1	S	C3	P3
9	How to change the RedHat Linux target in the current session?	1	S	C3	P3
1	What are the various types of System Units? Give the significance of any three	7	K	C2	P4
2	What is systemd ? What are the features of systemd ?	7	K	C1	P4
3	What is httpd? How to install httpd using package manager in RedHat Linux?	7	K	C1	P4
4	Give the difference between systemctl enable httpd.service and systemctl start httpd.service	7	U	C2	P4
5	Give the difference between systemctl restart httpd.service and systemctl try-restart httpd.service .	7	U	C2	P3
6	What is system targets in RedHat Linux? Name the different target units with its significance	7	U	C3	P3
7	What is the significance of systemctl reload httpd.service command? How it differs from systemctl restart httpd.service	7	S	C3	P3
8	How to find default target unit in RedHat Linux? How to change it to another target unit?	7	S	C3	P3
9	What is the significance of masking a service? Give the procedure to mask a specific service	7	S	C3	P3
10	Demonstrate how systemd manages various services in RedHat Linux	7	S	C3	P3
1	What is the significance of suspend and hibernate in case of RedHat Enterprise server? Demonstrate its usage.	10	S	C1	P4
2	What is systemd ? Give the procedure to manage the remote machine using systemd	10	S	C3	P3

Module 5

Sl No	Question	Marks	Type	CO	PO
1	SSH stands for	1	K	C1	P4
2	What is a Cipher Text?	1	K	C2	P4
3	Name the web server in RedHat Linux	1	K	C2	P4
4	What is Encryption?	1	U	C2	P4
5	What is Impersonation attack in cyber space?	1	U	C2	P4
6	What do you mean by IP Spoofing?	1	U	C2	P4
7	How to set web server to run by default in RedHat Linux?	1	S	C3	P3
8	How to install Apache web server in RedHat Linux?	1	S	C3	P3
9	How to install SSH server in RedHat Linux?	1	S	C3	P3
1	What is SSH? What are the main features of SSH?	7	K	C2	P4
2	What is sftp ? List the different commands with their significance.	7	K	C1	P4
3	What is SSL? Give its functionalities.	7	K	C1	P4
4	Give the steps involved in establishing a SSH communication between two hosts.	7	U	C2	P4
5	What is Virtual Hosting? Analyze different types of Virtual Hosting.	7	U	C2	P3
6	What are the packages required for having SSH communication? Explain	7	U	C3	P3
7	Give the procedure to start SSH service at a server and to log in to it from a remote PC.	7	S	C3	P3
8	What is scp ? Give the procedure to copy a local file to a remote file and also remote file to a local file using SSH	7	S	C3	P3
9	Give the procedure to set up a RedHat Linux Server as a web server.	7	S	C3	P3
1	What is Virtual Hosting? Give the procedure to set RedHat Linux Server for virtual hosting.	10	S	C1	P4
2	What is SSH server? Give the procedure to install and set SSH sever in a RedHat Linux	10	S	C3	P3

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